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Roll No

MI-5005(2)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Electronic Instrumentation

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss the basic principle working of FET.
b) Discuss the working and applications of SCR.
2. a) With circuit diagram, explain emitter stabilized biased circuit. Write the necessary equations.
b) An emitter bias circuit has $R_C = 2k\Omega$, $R_E = 680\Omega$, $V_E = 2.1V$, $V_{CE} = 7.3V$, $I_B = 20\mu A$. Find V_{CC} , R_B and β .
3. a) For a bipolar junction transistor compare between CB, CE and CC configuration.
b) Explain in detail how transistor works as an amplifier.
4. Explain MOSFET in following points
 - i) Construction
 - ii) Operation
 - iii) Circuit symbol
 - iv) Characteristics

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5.
 - a) Discuss the principle working of photo transistors.
 - b) Discuss about static and dynamic transistor characteristics.
6.
 - a) Derive an expression for the stability factor $S(I_{CO})$ of a voltage divider bias circuit.
 - b) What is biasing? Discuss the factors causes for bias instability in a transistor.
7. Explain FET small signal model with the help of graphical representation of g_m .

OR

- a) Discuss about transistor stabilization against variation of I_{CO} , V_{BE} and β .
 - b) Discuss about thermal runaway and stability.
8. Write short notes on any two of the following:
 - a) Voltage doubler
 - b) Clipper and Clampers
 - c) Darlington pair
